

Bayesian Reasoning: Beliefs and Evidence

Pinker's Chapter on Rationality

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The Core of Bayesian Reasoning

How do we update our beliefs when we encounter new evidence?

- **Prior Probability:** Our initial credence in a hypothesis (how plausible it is based on past experience).
- **Likelihood:** How likely the evidence would be if the hypothesis were true.
- **Marginal Probability:** How common the evidence is across the board (true and false instances combined).
- **Posterior Probability:** Our updated belief after weighing the prior against the new evidence.

The Representativeness Heuristic & Base-Rate Neglect

- **The Trap:** We judge probabilities by how closely a subject matches a stereotype, completely ignoring statistical base rates.
- **The Conjunction Fallacy:** Assuming "Linda the feminist bank teller" is more likely than "Linda the bank teller" because of narrative fit.
- **Expert Errors:** Doctors, ecologists, and taxonomists often flub diagnostic probabilities by overweighting a positive test result and forgetting how rare the condition is in the overall population.

Why extraordinary claims require extraordinary evidence

- **The Replicability Crisis:** Journals heavily favor "surprising" (low prior) discoveries. If the prior is infinitesimal, even a statistically significant result will likely fail to replicate.
- **Hume on Miracles:** The prior probability of a miracle is practically zero. The likelihood of false human testimony (lies, errors) is always mathematically higher.
- **Sagan's Standard:** Peer review must demand evidence (likelihood) strong enough to overpower shockingly low priors.

The Ethics of "Forbidden Base Rates"

When is it rational to ignore the math?

- **Systemic Justice:** Using demographic base rates (race, gender) for actuarial accuracy can create self-fulfilling prophecies and lock in historical disadvantages.
- **Redefining Rationality:** Rationality is applying knowledge to attain a goal. If fairness and public trust are higher goals than predictive accuracy, banning demographic profiling is entirely rational.
- **The Exception - Insurance:** Public policy must balance this. Forbidding all base rates in insurance would cause systemic collapse, forcing safe populations to subsidize reckless ones.

The Illusion of Perfect Prediction

Why we cannot profile rare events

- **Predicting the Impossible:** The public demands profiles to catch school shooters or terrorists before they act.
- **The Math of False Positives:** Because the base rate of terrorists in the population is microscopic, even a 99 percent accurate predictive test will mostly finger the innocent.
- **The Reference Class Problem:** If we try to narrow a reference class to establish a perfect prior for an individual, the sample size becomes so small that the estimate becomes noisy and useless.

Bridging the gap between brains and statistics

- **Natural Frequencies:** Humans struggle with abstract conditional probabilities (decimal fractions). We evolved to count discrete events.
- **The Solution:** Reframe "90 percent sensitivity and 9 percent false alarms" as "9 true positives and 89 false positives out of 1000."
- **Visual Arrays:** Transforming data into visual arrays of squares allows our primate visual cortex to instantly compare the proportion of true vs. false outcomes (e.g., the taxicab problem).

Calibrating expectations in a competitive world

- Rejection in academia, hiring, or publishing often feels like a personal failure.
- A statistical understanding of base rates—the sheer volume of highly qualified competitors vying for singular slots—reminds us of the mathematical reality.
- Remembering the base rate removes the sting of rejection, providing emotional equanimity and grounding our hopes in objective probability.